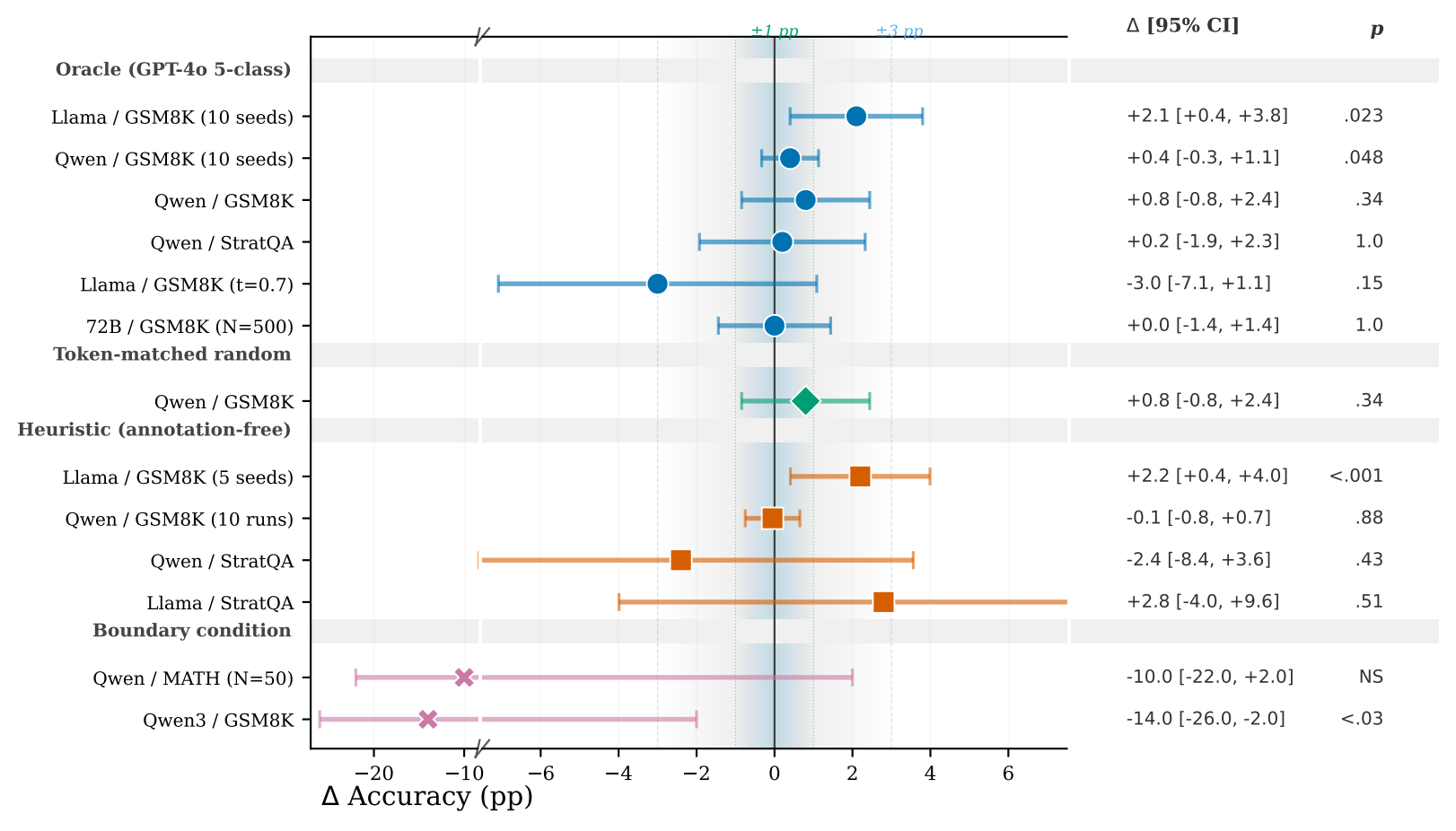
+1 pp
+3 pp
Δ [95% CI]
p
Oracle (GPT-4o 5-class)
Llama / GSM8K (10 seeds)
+2.1 [+0.4, +3.8]
.023
Qwen / GSM8K (10 seeds)
+0.4 [-0.3, +1.1]
.048
Qwen / GSM8K
+0.8 [-0.8, +2.4]
.34
Qwen / StratQA
+0.2 [-1.9, +2.3]
1.0
Llama / GSM8K (t=0.7)
-3.0 [-7.1, +1.1]
.15
72B / GSM8K (N=500)
+0.0 [-1.4, +1.4]
1.0
Token-matched random
Qwen / GSM8K
+0.8 [-0.8, +2.4]
.34
Heuristic (annotation-free)
Llama / GSM8K (5 seeds)
+2.2 [+0.4, +4.0]
<.001
Qwen / GSM8K (10 runs)
-0.1 [-0.8, +0.7]
.88
Qwen / StratQA
-2.4 [-8.4, +3.6]
.43
Llama / StratQA
+2.8 [-4.0, +9.6]
.51
Boundary condition
Qwen / MATH (N=50)
-10.0 [-22.0, +2.0]
NS
Qwen3 / GSM8K
-14.0 [-26.0, -2.0]
<.03
−20
−10
−6
−4
−2
0
2
4
6
Δ Accuracy (pp)